

DEVICE ANATOMY 2.0

Engineering Investigation of an Electronic Device at End-of-Life

Course	E-Waste & Environmental Management	Class	TE INFT
Duration	45-60 minutes	Marks	10

10
10

Group Details

Group No.:	11	Device Investigated:	DVD player
Date:	02 Sept 2026	Condition:	Non-functional.
Member 1:	Vishal R. Prajapati (24101B0047)	Member 2:	Meet M Alshi (24101B0048)
Member 3:	Altamash Ansari (24101B0048)	Member 4:	Pranitha Shetty (24101B0027)
Member 5:			

Condition: ☐ Functional ☒ Non-functional ☐ Damaged ☐ Unknown

Activity Evidence

During the activity, photograph all group members actively participating in the Device Anatomy investigation. Print the photograph and attach it below. A posed group selfie is not considered activity evidence.

ATTACH PRINTED GROUP PHOTOGRAPH HERE

BACK SIDE.

Photo caption: Our group investigating the anatomy and end-of-life characteristics of a DVD player.
Group members visible: Vishal, Meet, Pranitha, Altamash.

ACTIVITY CHALLENGE

You are an e-waste engineer. Your task is to investigate the material anatomy of an electronic product, identify recovery opportunities and environmental risks, and propose engineering changes that improve its end-of-life management.

Core question: Can you make this device more circular?

Part A — Device Profile

Device name	DVD Player
Primary function	To play DVDs, video CDs. to display video on it's built-in LCD screen.
Why does this device become e-waste?	☐ Obsolescence ☐ Damage ☐ Upgrade ☒ Repair not economical ☐ Failure ☐ Other: _____

Part B — Engineering Anatomy

Identify at least 6 major components. Do not merely name the component — explain the material choice.

#	Component	Function	Material(s)	Why is this material suitable?
1	LCD display	Display Video and UI	Glass, polymers, semiconductors	It provide a clear display surface to enable display control.
2	DVD Optical Drive	Reads data from DVDs/CD using Laser	Plastics, metals, electronic components	Lightweight and durability
3	Optical Pickup Laser	Reads information stored on the disc	Semiconductor + plastic material	Highly accurate reading of data from the disc
4	PCB	Controls processing & power	Copper, semiconductor	Good electronic processing and control
5	Speakers	Converts electrical signal into sound	Plastic, rubber	lightweight, insulation and flexibility
6	Plastic Housing	Internal components	Engineering Plastic	Durable, electrically insulation

Engineering Challenge: Which component demonstrates the most interesting material selection? Why?

LCD display because it combines glass, polymers and semiconductor materials to produce a clear image while keeping the device lightweight and durable

Part C — Material Detective

Tick the major material categories present in your device:

- ☐ Ferrous metals ☒ Aluminium ☒ Copper ☐ Precious metals
☒ Engineering plastics ☒ Glass ☒ Ceramics ☒ Semiconductor materials
☒ Rubber/elastomers ☐ Other: _____

	Material 1	Material 2
Name	Copper	Engineering plastic
Where is it used?	Power cables, PCB tracks, connectors & internal wiring	outer casing, buttons, disc, internal support
Why is it used?	Because copper conducts electricity	Its lightweight, strong and inexpensive

	Material 1	Material 2
Can it be recovered/recycled? How?	Yes, Copper parts can be separated during dismantling	Yes, after separating different plastic types.

Part D — Hidden Value: The Urban Mine

Identify the three most valuable/recoverable materials in your device.

Material	Component containing it	Potential value	Recovery difficulty
1. Copper	PCB, wires, speakers & connectors	Low / Medium / <u>High</u>	Low / <u>Medium</u> / High
2. Aluminium	Heat dissipation parts & electronic components	<u>Low</u> / Medium / High	Low / Medium / <u>High</u>
3. Precious metals	PCB & electronic contacts	Low / Medium / <u>High</u>	Low / Medium / <u>High</u>

Critical Thinking: Is the most valuable material necessarily the easiest to recover? Explain.

No. Precious metals may have high value but are present in very small quantities and are embedded in electronic components.

Part E — Hazard Detective

Identify components that may require controlled handling.

Component	Possible hazard	Risk if improperly disposed	Handling requirement
Rechargeable battery	Can leak, short circuit or be damaged	Fire risk.	Controlled recycling
PCB	Contains various electronic materials.	Pollution & resource loss	E-waste recycling
LCD screen	Contains glass & electronic layers	Can break & create difficult	Handle carefully.
Plastic casing	Improper burning & harmful fumes	Air pollution	Recycle, do not burn.

Most environmentally concerning component: Rechargeable battery

Why? The battery contains electrochemical materials and must be handled separately. Improper disposal can cause leakage, fire hazards and environmental contamination.

Part F — End-of-Life Decision

Recommend the most appropriate pathway for each major component.

Component	Reuse	Repair	Refurbish	Recycle	Controlled Disposal
LED Display	☐	☐	☐	☒	☐
DVD Drive	☐	☐	☒	☐	☐
PCB	☐	☐	☒	☐	☐
Copper wires	☐	☐	☐	☒	☐

Justify ONE of your decisions: Battery → Controlled Disposal: Recycling.
The battery should not be thrown into normal household waste because it contains chemical and metallic materials.

Part G — Design for Disassembly

Imagine that you are the engineer who designed this product. Identify three design features that make recycling, repair or disassembly difficult.

#	Existing design problem
Problem 1	The outer casing is difficult to open for accessing internal components.
Problem 2	The battery may be difficult to remove or replace.
Problem 3	Different materials and electronic modules are tightly packed together.

Now redesign it!

Propose three engineering modifications that would make the product more circular.

Existing Design Problem	Proposed Design Change	Sustainability Benefit
The Outer Design	Standard screws instead of permanent clips	Makes repair and dismantling easier
Battery	Replaceable battery module with accessible connector	Failed battery would not make device unusable
Electric modules	Modular internal design	Component replacement & material recovery

Sketch your redesigned device/component — show at least one change that improves repair, disassembly or material recovery.

ORIGINAL	REDESIGNED
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LCD Screen DVD Drive PCB + Battery	LCD Screen DVD Drive Battery Module (Removable PCB)	Adj: Screw-open casing Replaceable battery Modular component Easy recycling
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Problem:- Glued casing
Fixed Battery
Difficult repair

Part H — E-Waste Journey

Trace the end-of-life journey of ONE component.

Component selected: Rechargeable Battery

DEVICE	CONSUMER DISCARD	COLLECTION	DISMANTLING	MATERIAL RECOVERY	NEW PRODUCT / RESOURCE
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What could go wrong at any stage?

If the battery is thrown into regular waste, it may be damaged, leak or create a fire hazard, while valuable materials are also lost.

Part I — Circular Economy Decision

Our device should NOT simply be treated as waste because... It contains valuable materials and components such as copper, aluminium, plastics, electronic components etc. They can potentially be repaired and, recovered or reused.

Which circular strategy would have the greatest impact?

☐ Repair ☐ Reuse ☒ Refurbishment ☐ Remanufacturing

☐ Recycling ☐ Material recovery

Why?

A portable DVD player can potentially be refurbished by replacing the battery, optical drive or damaged components while keeping the main body in use. This can significantly extend its useful life and reduce e-waste.

Part J — Engineering Viva

Be prepared to defend your answers. Your instructor may ask any of the following:

1. Why was a particular material selected for one component? → Cu
2. Which material has the greatest recovery potential? → Cu
3. Which component is most difficult to recycle and why? → PCB
4. What is the single most effective design modification you propose?
5. If your sustainable redesign increases manufacturing cost by 8%, would you still recommend it? Why?

Final Group Verdict

The most valuable component is:	PCB and copper containing component
The component with the greatest environmental concern is:	Rechargeable battery
The biggest design-for-recycling problem is:	Difficult access to and separation of internal components
Our most important redesign recommendation is:	Use a screw-open modular design with replaceable battery.
ONE-SENTENCE ENGINEERING VERDICT We recommend <u>redesigning the portable DVD player</u> because <u>it's useful to reduce e-waste</u>	

Assessment Rubric — 10 Marks

Criterion	Excellent (2)	Good (1.5)	Developing (1)	Marks
1. Engineering Anatomy	Accurate component + function + material analysis	Mostly accurate	Basic identification	/2
2. Material & Recovery Analysis	Strong material/recovery reasoning	Adequate reasoning	Limited reasoning	/2
3. Hazard & E-waste Assessment	Identifies risks with clear justification	Identifies major risks	Basic identification	/2
4. Design for Disassembly	Innovative, feasible engineering solutions	Relevant solutions	Basic suggestions	/2
5. Engineering Defence/Viva	Clear, evidence-based justification	Reasonable explanation	Limited justification	/2
TOTAL				10

Reflection

What surprised you most about the materials hidden inside your device?

I was surprised that small portable DVD player contains many different materials and components.

After this activity, would you still call this device "waste"? Why?

No, because many of its components and materials can potentially be reused, repaired or recovered.

KEY TAKEAWAY

E-waste is not merely waste — it is a complex mixture of materials, components, resources and risks that requires engineering solutions for responsible recovery.

